

Basic Principles of Medicine 1

Module: Musculoskeletal System

Lecture No: 23

Neuromuscular Transmission and Excitation-contraction coupling

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Objectives

- SOM.MK.II.BPM1.2.MSK.2.PHYS.0052 Explain the overall transmembrane signaling steps whereby increases in cytosolic calcium initiate crossbridge cycling
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0053 Identify the multiple sources, localization, and roles of calcium in muscle contraction and relaxation.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0054 Illustrate the chemical and mechanical steps in the cross-bridge cycle and explain how the cross-bridge cycle results in shortening of the muscle.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0055 Explain the relationship of myosin-thick filament bare zone (I zone) to the shape of the active length: force relationship.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0056 List the steps in excitation contraction coupling in skeletal muscle, and describe the roles of the sarcolemma, transverse tubules, sarcoplasmic reticulum, thin filaments, and calcium ions.

Objectives

- SOM.MK.II.BPM1.2.MSK.2.PHYS.0057. Describe the roles of ATP in skeletal muscle contraction and relaxation.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0058 List in sequence the steps involved in neuromuscular transmission in skeletal muscle and point out the location of each step on a diagram of the neuromuscular junction.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0059 Distinguish between an endplate potential and an action potential in skeletal muscle.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0060. List the possible sites for blocking neuromuscular transmission in skeletal muscle and provide an example of an agent that could cause blockage at each site.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0061 Describe the role of the myosin cross bridges acting in parallel to determine active force.

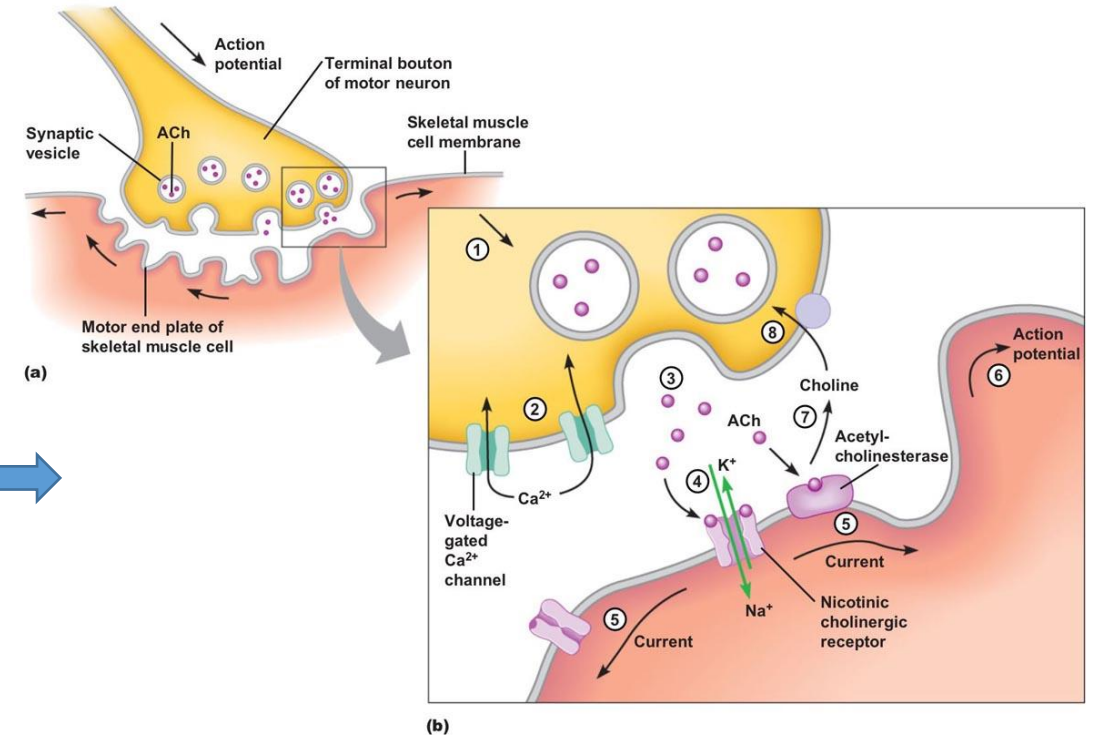
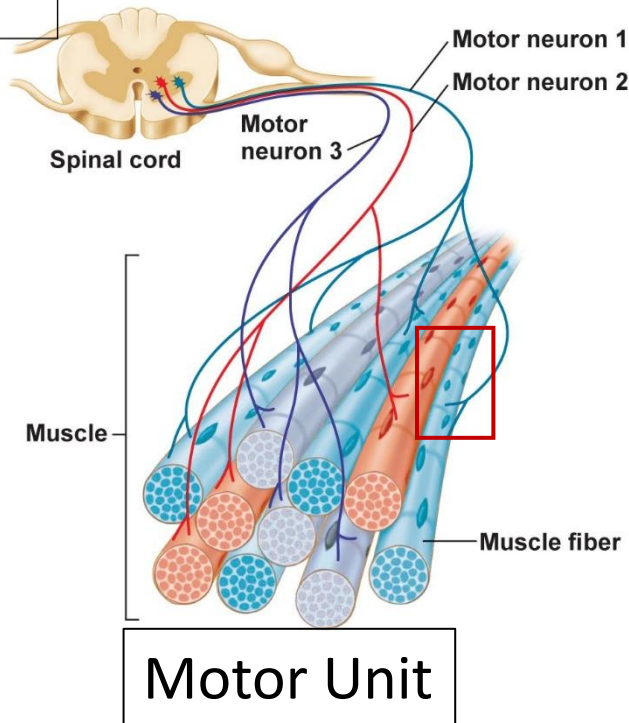
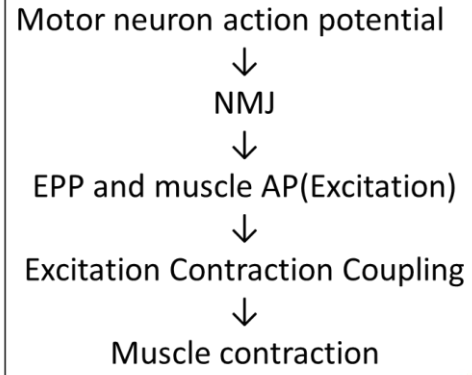
Recommended Readings

Rhoades, R. & Bell, D. (2023); *Medical Physiology: Principles for Clinical Medicine*, 6th Edition; Wolters Kluwer. Chapter 25

(Slides with blue backgrounds are not included in the lecture time but included in all assessments)

The Big Picture: Neuromuscular system

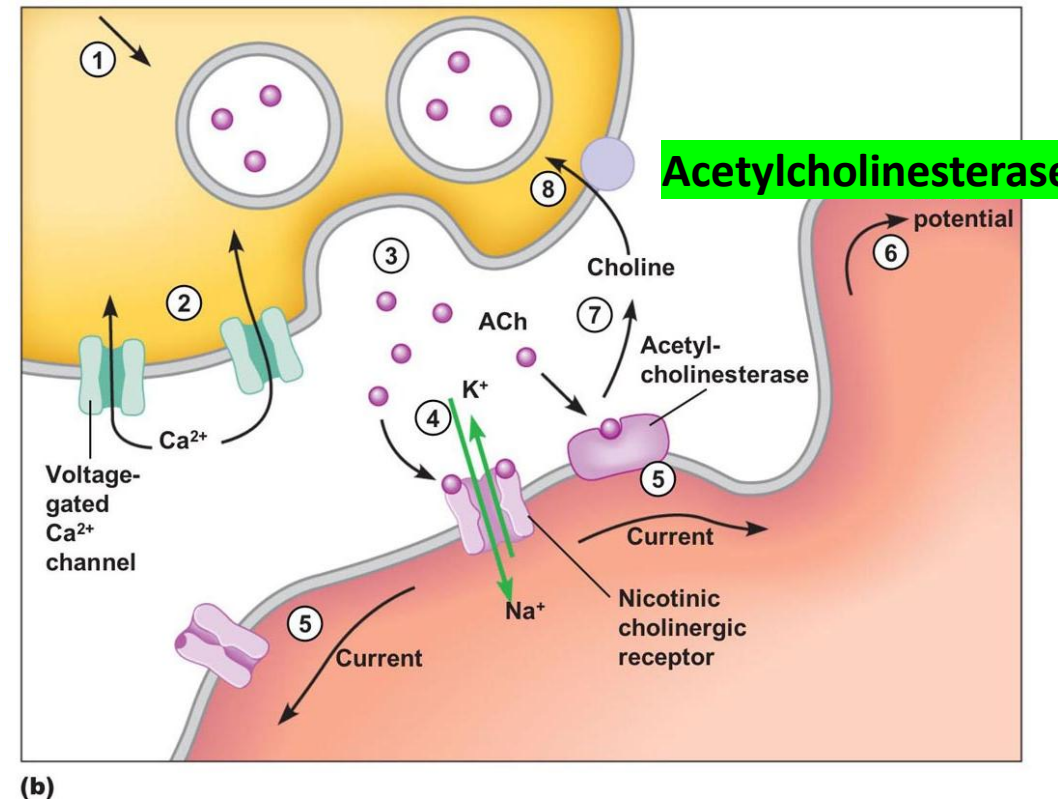
Sequence of events



Synapse between a motor neuron and a muscle fiber

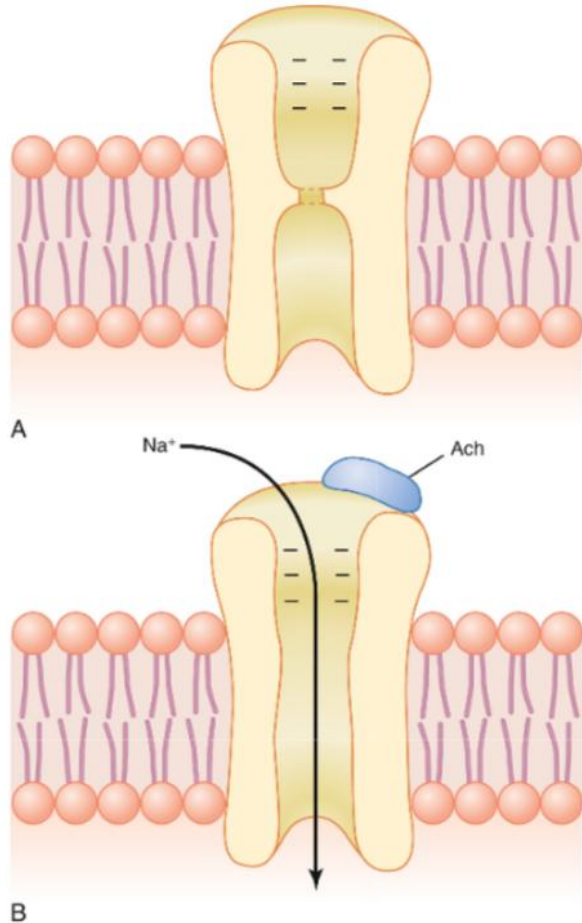
Communication at the NMJ

1. Action potential arrives at terminal bouton
2. Voltage-gated calcium channels open
3. Calcium enters cell triggering release of ACh
4. ACh diffuses across cleft and binds to nicotinic receptors on motor end plate
5. ACh triggers opening of channels for small cations sodium and potassium
6. **Net movement of positive charge in → depolarization**
 - **End plate potential = EPP**
7. EPP causes action potential in muscle cell
8. Action potential spreads through muscle causing contraction



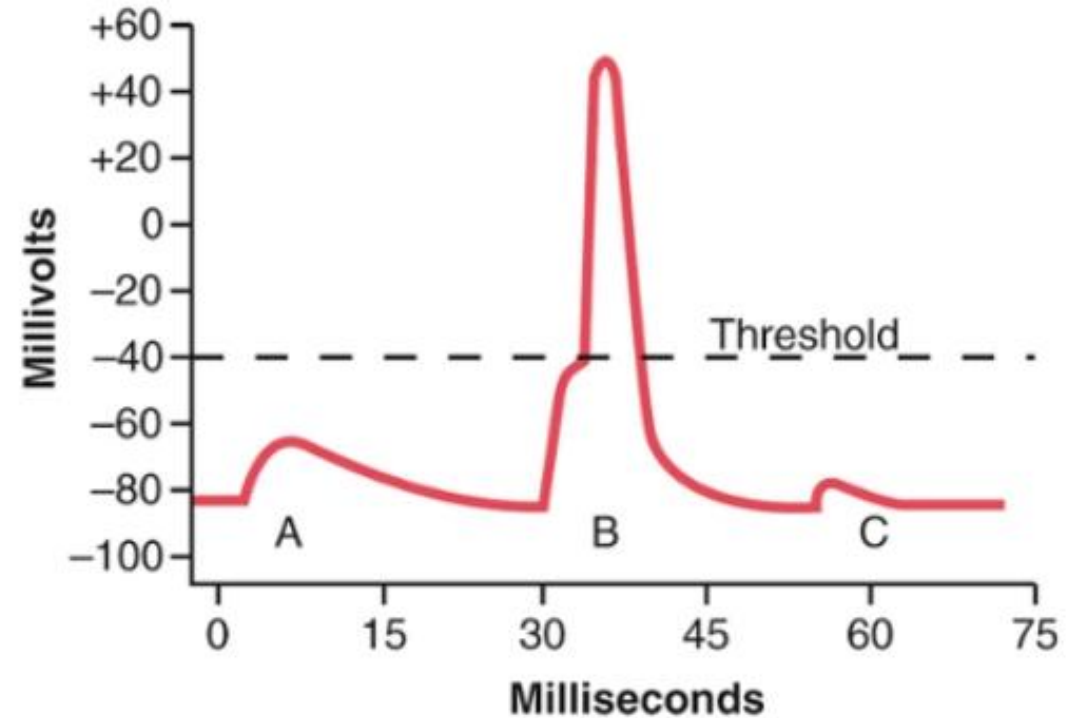
EPP: End plate potential

What is so special about EPP?



Motor endplate

Acetylcholine-gated channel



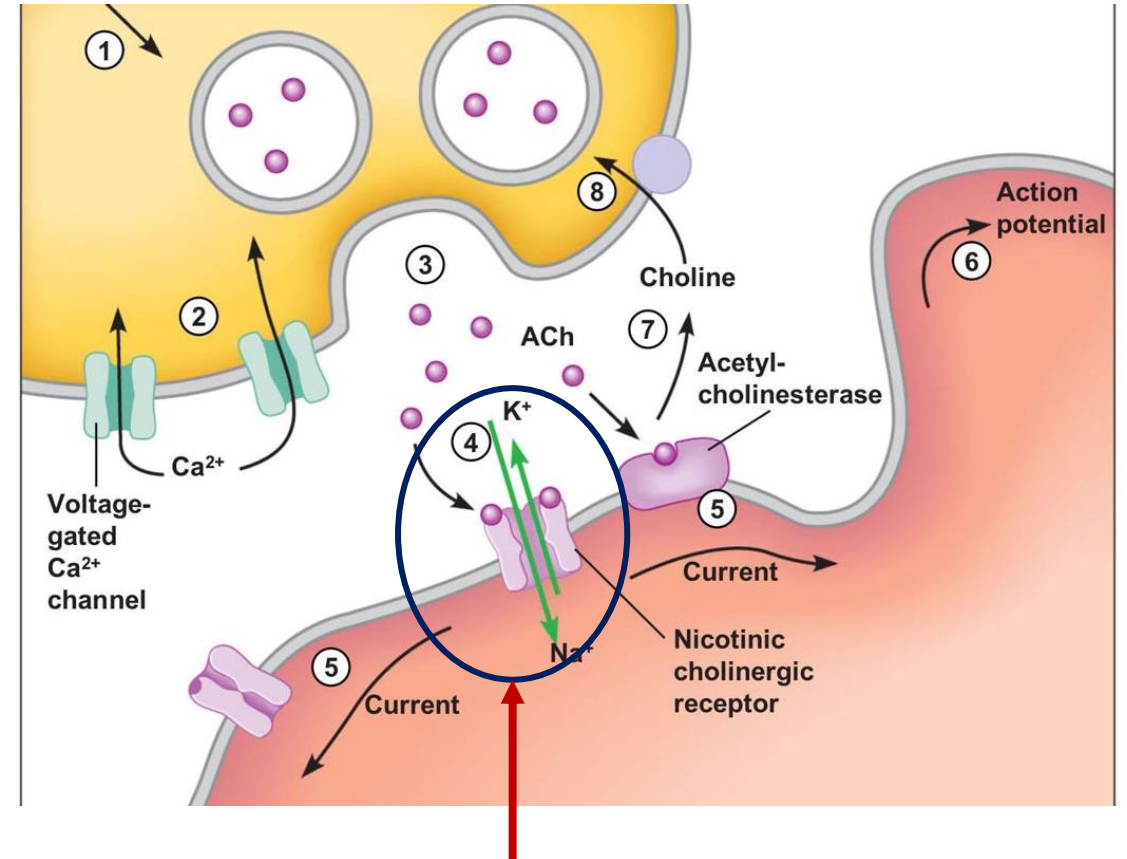
- A. Below threshold, Weak, not propagate
- B. Above threshold, Muscle AP, propagate

NMJ Clinical Correlations

- Weakness and fatigue of the skeletal muscles. In one form, the extraocular muscles are primarily affected.
- Caused by the formation of circulating antibodies to the nicotinic acetylcholine receptors.
- Weakness improves after a period of rest or after administration of **acetylcholinesterase inhibitors**.

Tensilon Test

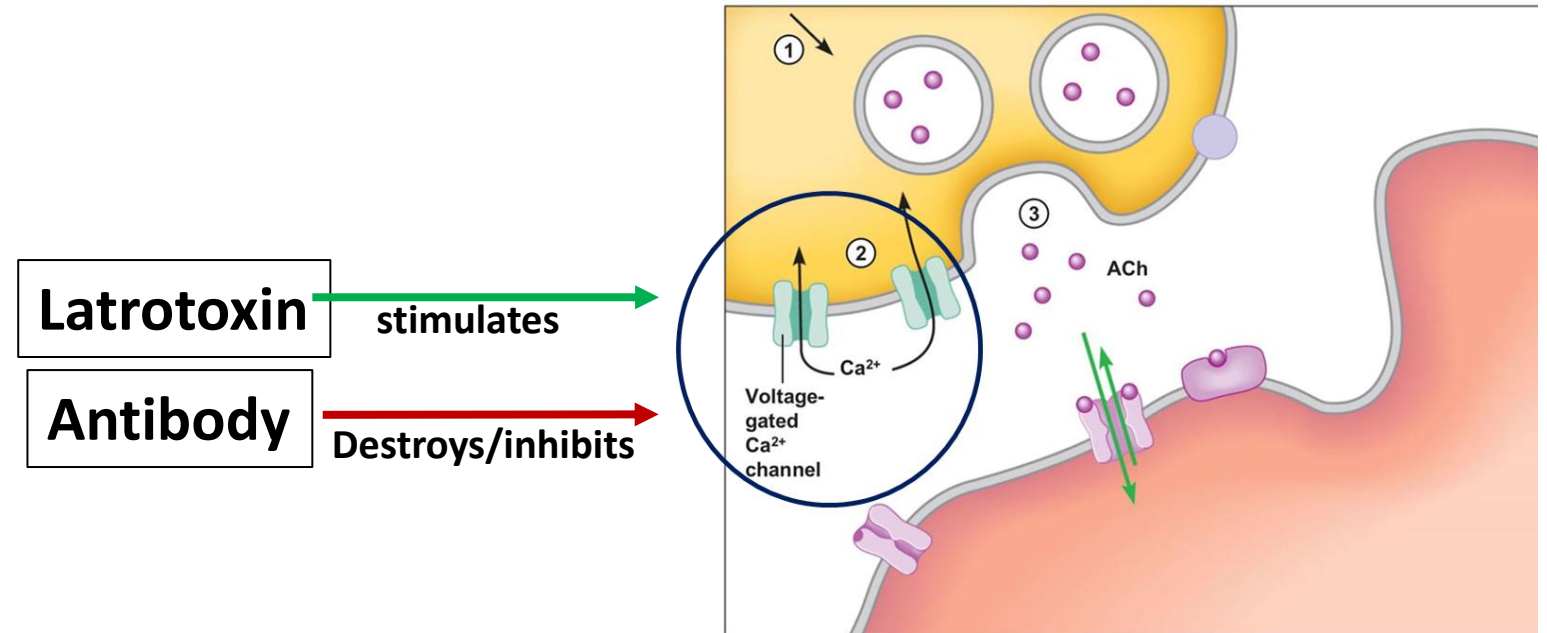
Myasthenia Gravis



Autoimmune damage of
nicotinic cholinergic receptors

Clinical Correlation of Lambert–Eaton Myasthenic Syndrome (LEMS) and Latrotoxin Poisoning

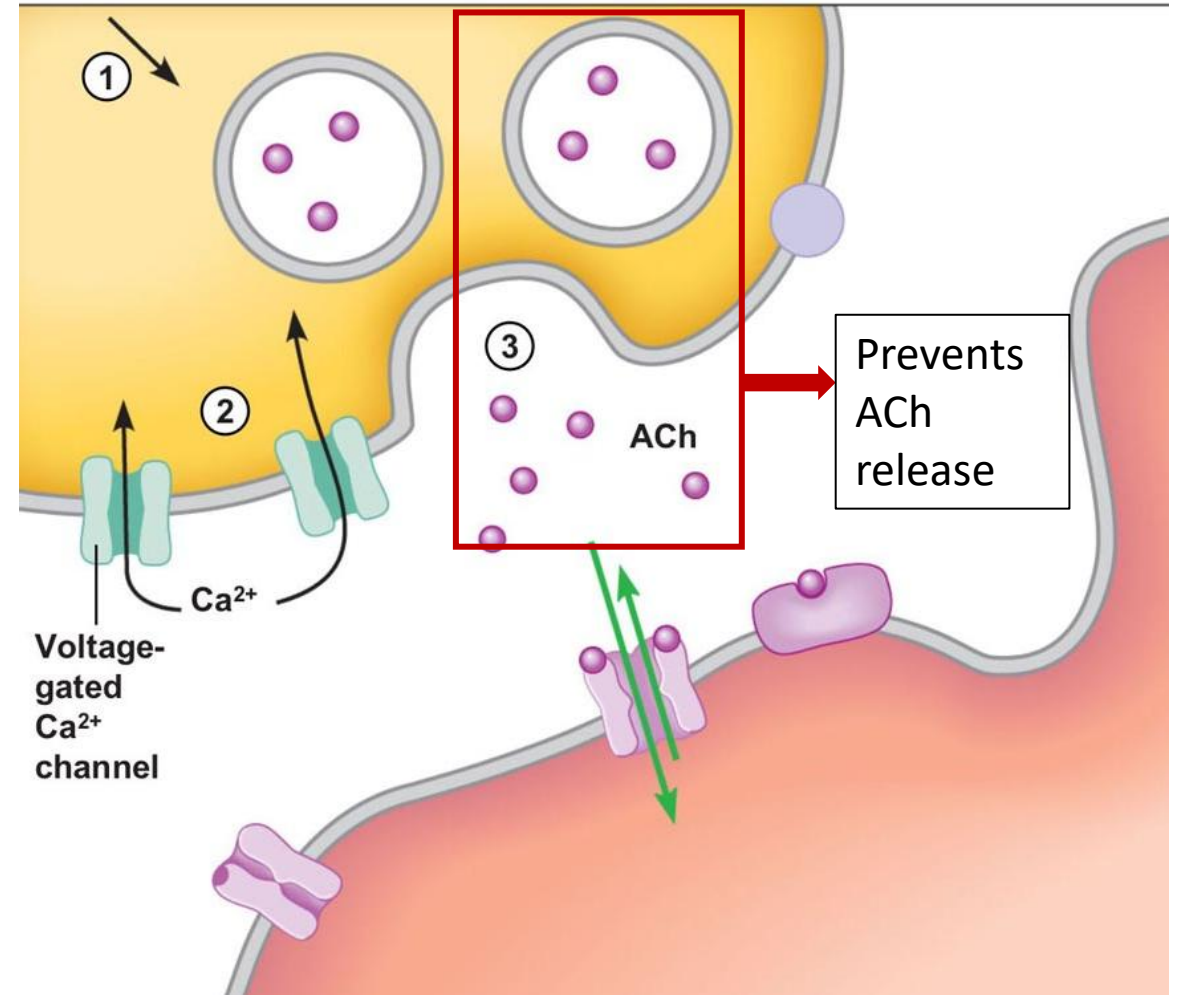
- Muscle weakness is caused by an autoimmune attack against Ca^{2+} channels in the nerve endings at the NMJ.
- $\downarrow \text{Ca}^{2+}$ influx at NMJ $\rightarrow$ $\downarrow$ ACh release.
- Proximal muscles of the lower extremities are primarily affected, producing a waddling gait and difficulty raising the arms.
- Difference with myasthenia gravis



Latrotoxin: The venom of the black widow spider, it opens presynaptic Ca^{2+} channels resulting in excess ACh release $\rightarrow$ profound muscle contraction $\rightarrow$ muscle spasms, muscle pain, and abdominal rigidity

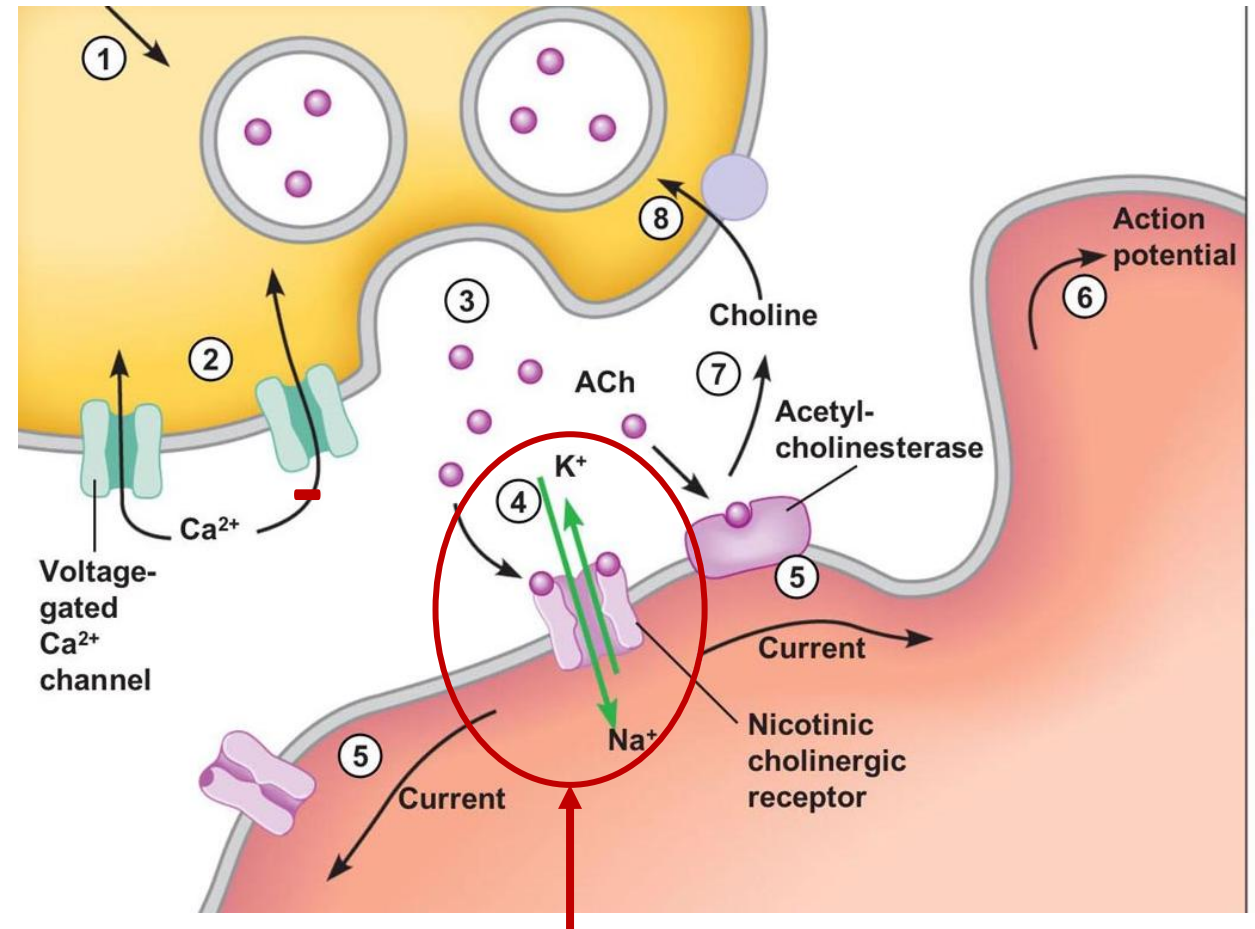
Botulinum toxin/ Botulism/Botox

- Produced by the bacterium *Clostridium botulinum*. Food poisoning by ingestion of this organism can cause **Botulism**.
- Protease that prevents release of ACh vesicles from the NMJ.
- Symptoms include weakness of muscles innervated by cranial nerves, blurred vision, flaccid Paralysis
- In small doses, it is used in cosmetic treatment of **Botox**



NMJ blockers / Muscle Relaxants

- Tubocurarine and is a N_m receptors antagonist and used as a skeletal muscle relaxant.
- Curare is a plant-derived alkaloid, used as arrows/blow dart poison. It is a competitive inhibitor of N_m

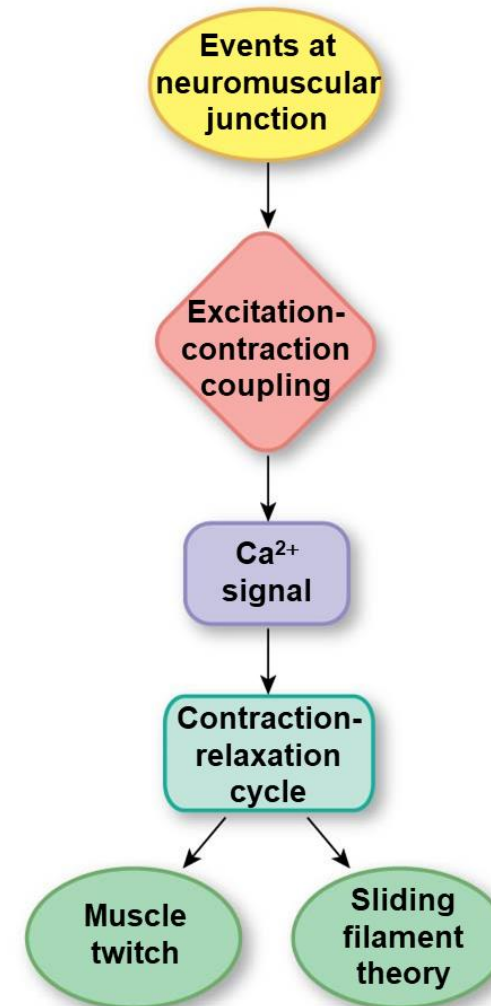


**Tubocurarine
Or Curare**

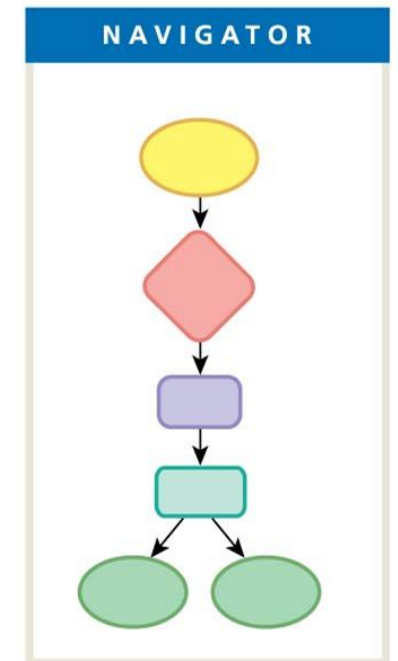
Muscle Contraction

- Contraction: Creation of tension in muscle
- Relaxation: Release of tension
- Muscle tension: Force created by muscle
- Load: Weight or force opposing contraction

SUMMARY MAP OF MUSCLE CONTRACTION



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This icon represents a map of the events in muscle contraction. Look for this icon throughout this chapter as these events are explored in greater detail.

Excitation Contraction Coupling

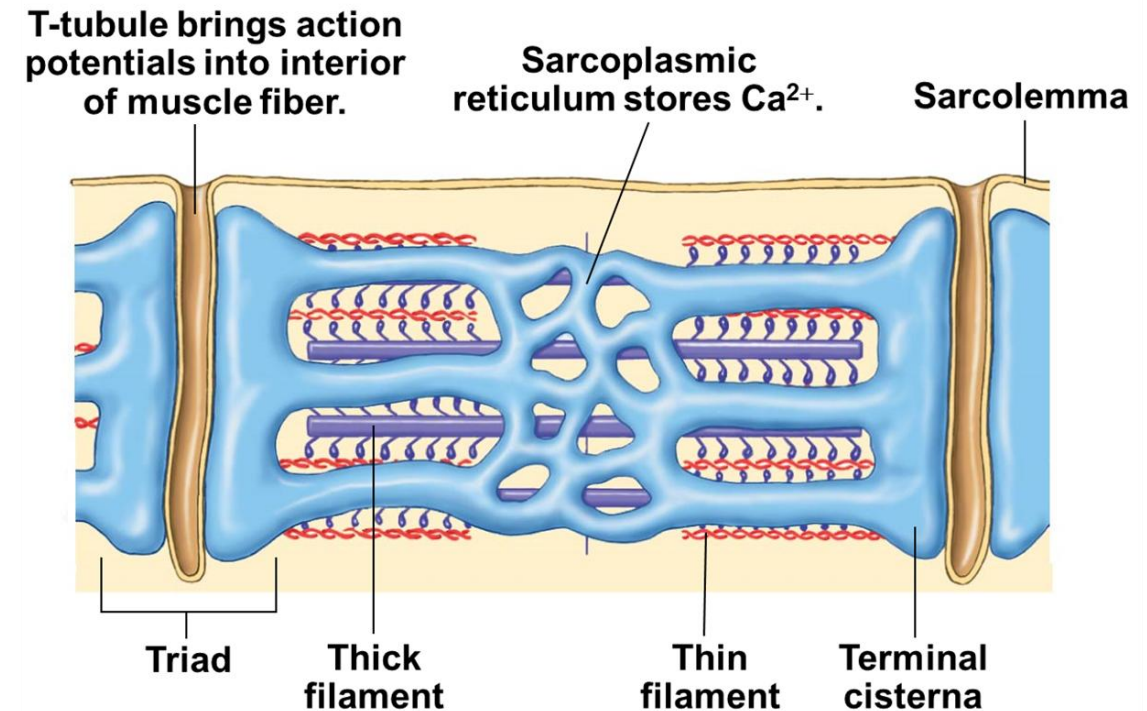
Resting muscle

- Intracellular $[Ca^{2+}]$ very low ($<10^{-9}M$ or nM).
- Majority Ca^{2+} stored in sarcoplasmic reticulum (SR).
- Ca^{2+} bound to SR proteins e.g., calsequestrin.

Contracting muscle

- Muscle depolarization $\rightarrow$ Ca^{2+} release from SR.
- SR ONLY source of Ca^{2+} for contraction (skeletal muscles)
- Intracellular $[Ca^{2+}]$ increased ($>10^{-6}M$ or μM)

Sarcoplasmic Reticulum (SR)



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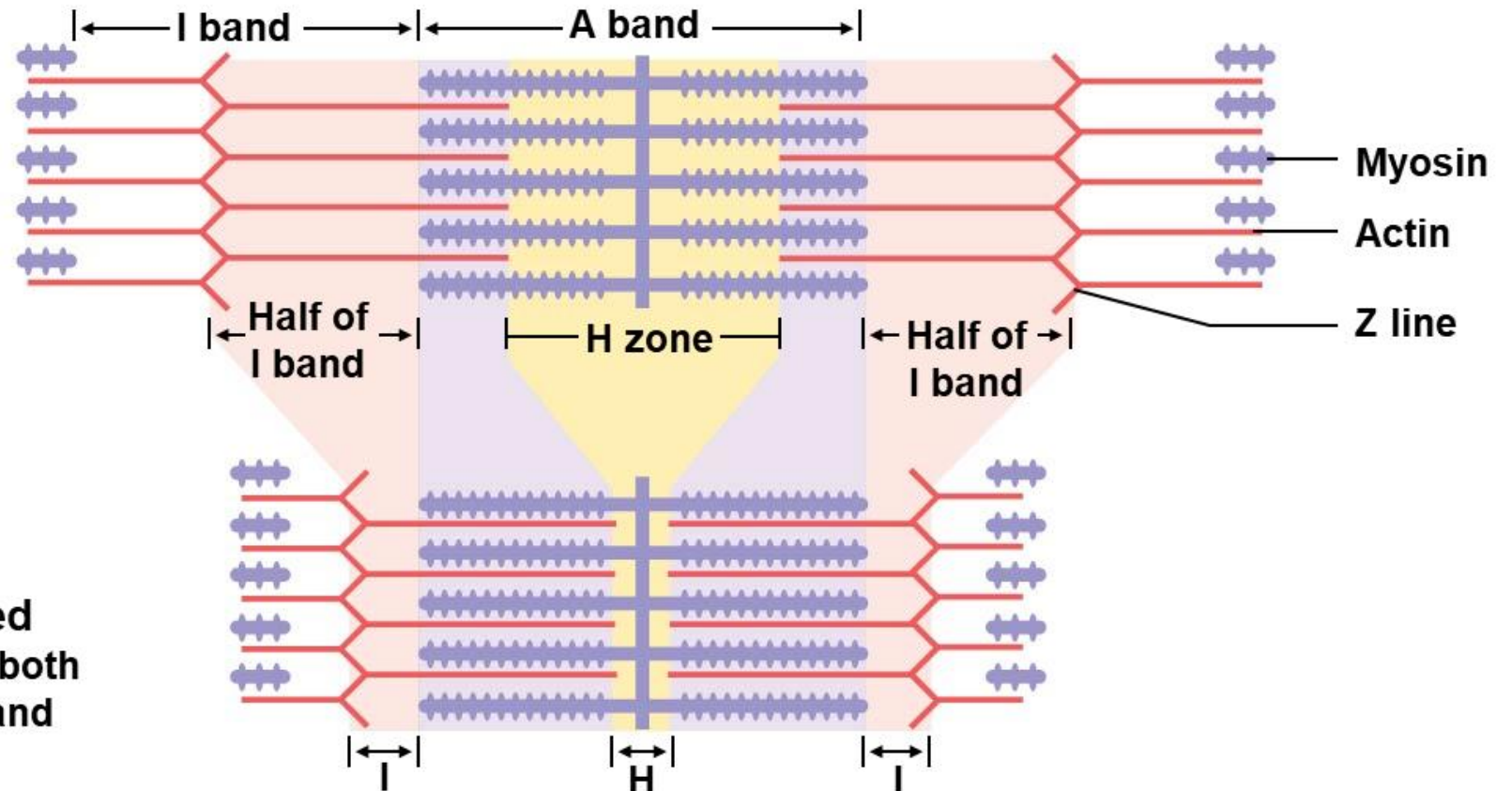
Sarcomere and Contraction

The Sarcomere Shortens During Contraction. As contraction takes place, actin and myosin do not change length but instead slide past one another.

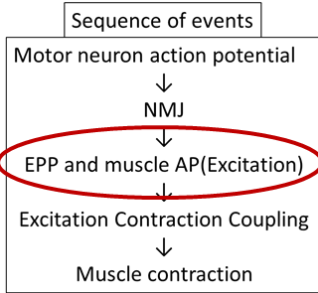
(d) Muscle Relaxed

Sarcomere shortens with contraction.

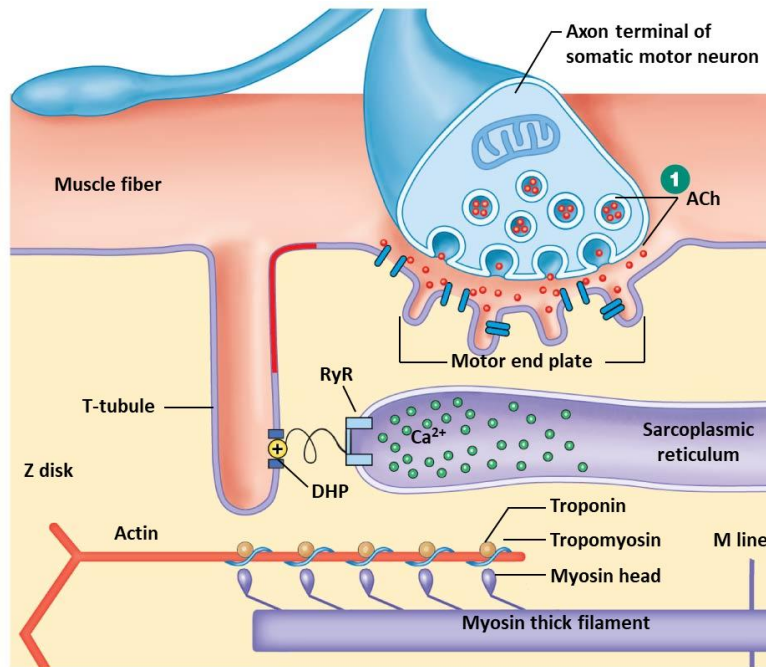
(e) Muscle Contracted
H zone and I band both shorten, while A band remains constant



Excitation-Contraction Coupling and Relaxation



(a) Initiation of Muscle Action Potential

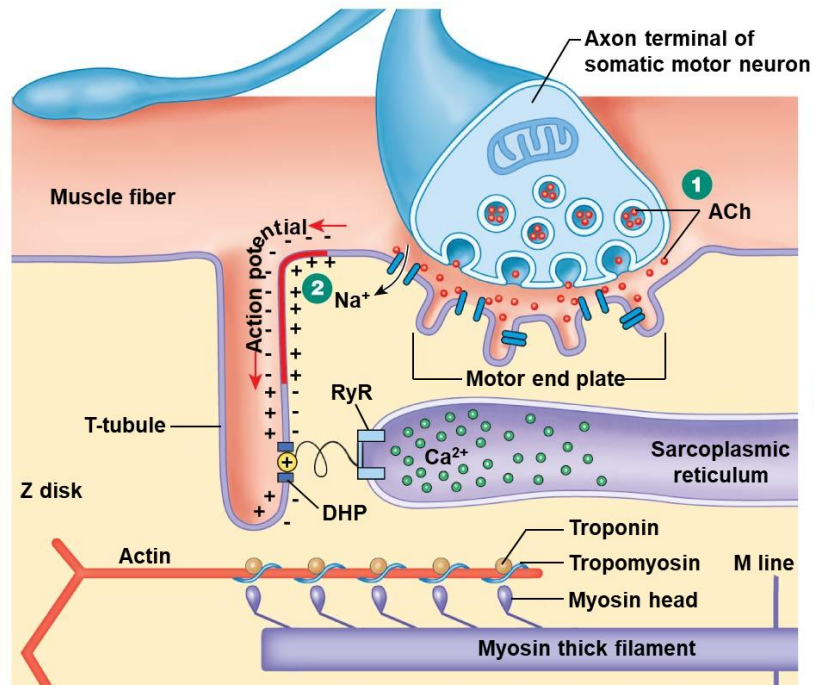


KEY

DHP = dihydropyridine L-type calcium channel

RyR = ryanodine receptor-channel

- 1 Somatic motor neuron releases ACh a neuromuscular junction.



KEY

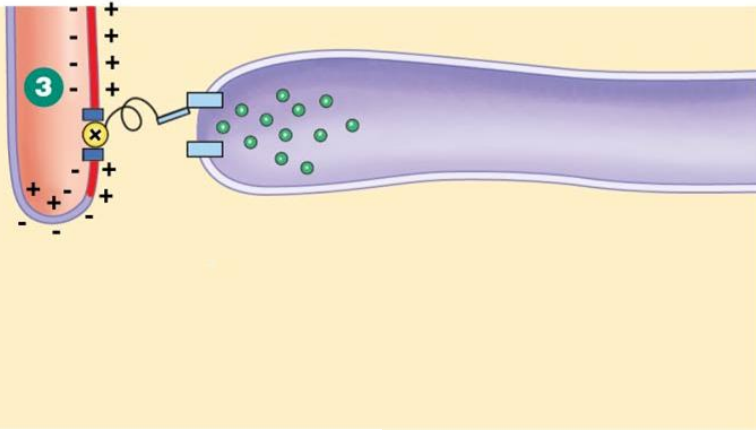
DHP = dihydropyridine L-type calcium channel

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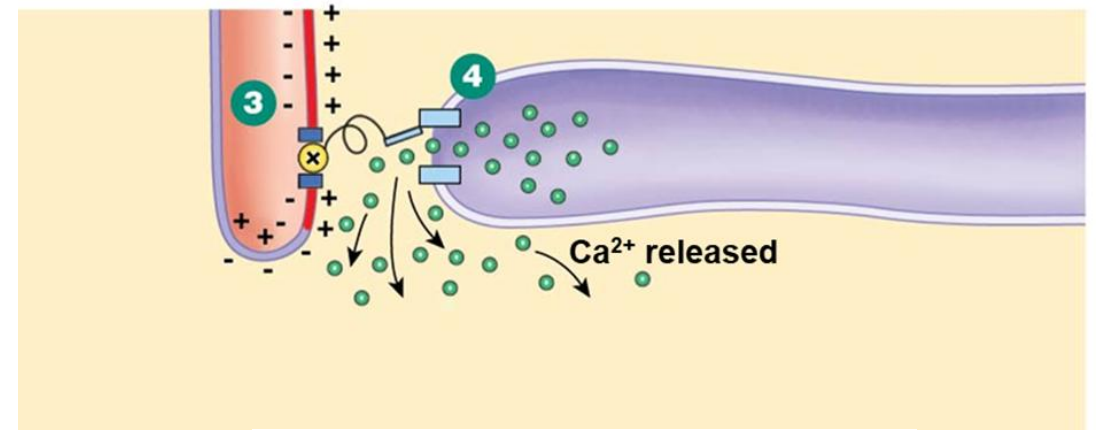
- 1 Somatic motor neuron releases ACh a neuromuscular junction.
- 2 Net entry of Na^+ through ACh receptor-channel initiates a muscle action potential.

Excitation-Contraction Coupling and Relaxation

(b) Excitation-Contraction Coupling



3 Action potential in t-tubule alters conformation of DHP receptor.



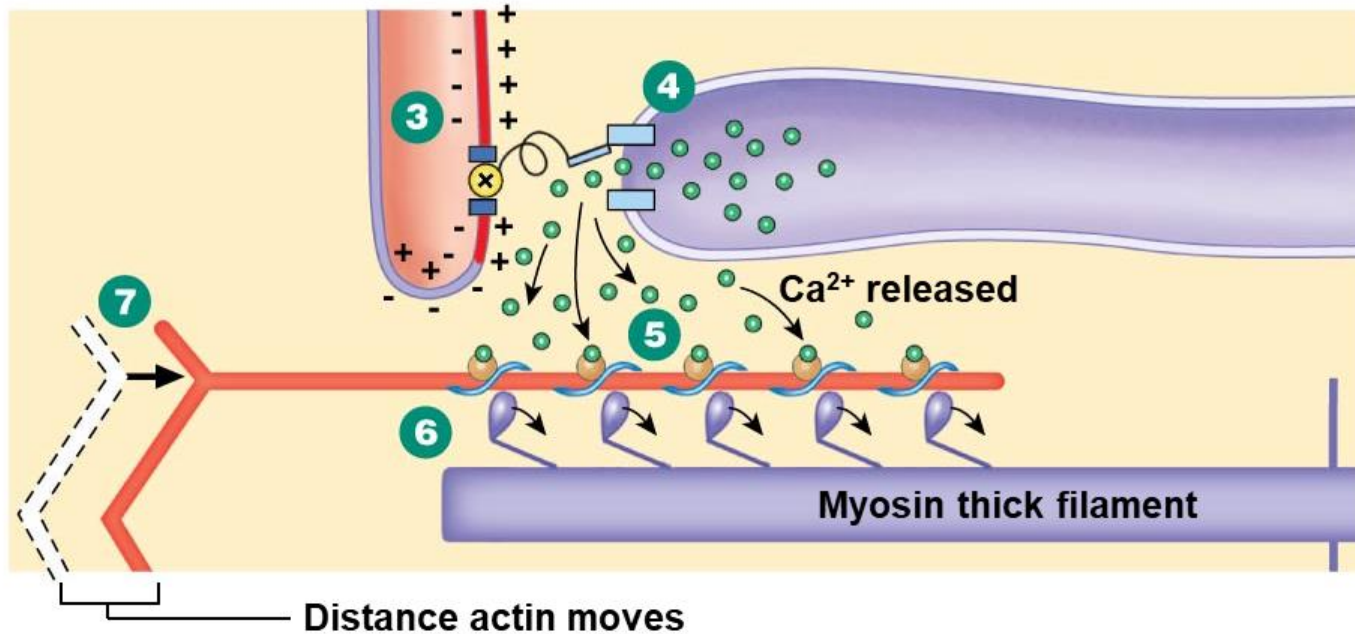
3 Action potential in t-tubule alters conformation of DHP receptor.

4 DHP receptor opens RyR Ca^{2+} release channels in sarcoplasmic reticulum, and Ca^{2+} enters cytoplasm.

DHP = dihydropyridine L-type calcium channel
RyR = ryanodine receptor-channel

Excitation-Contraction Coupling and Relaxation

(b) Excitation-Contraction Coupling



3 Action potential in t-tubule alters conformation of DHP receptor.

4 DHP receptor opens RyR Ca^{2+} release channels in sarcoplasmic reticulum, and Ca^{2+} enters cytoplasm.

5 Ca^{2+} binds to troponin, allowing actin-myosin binding.

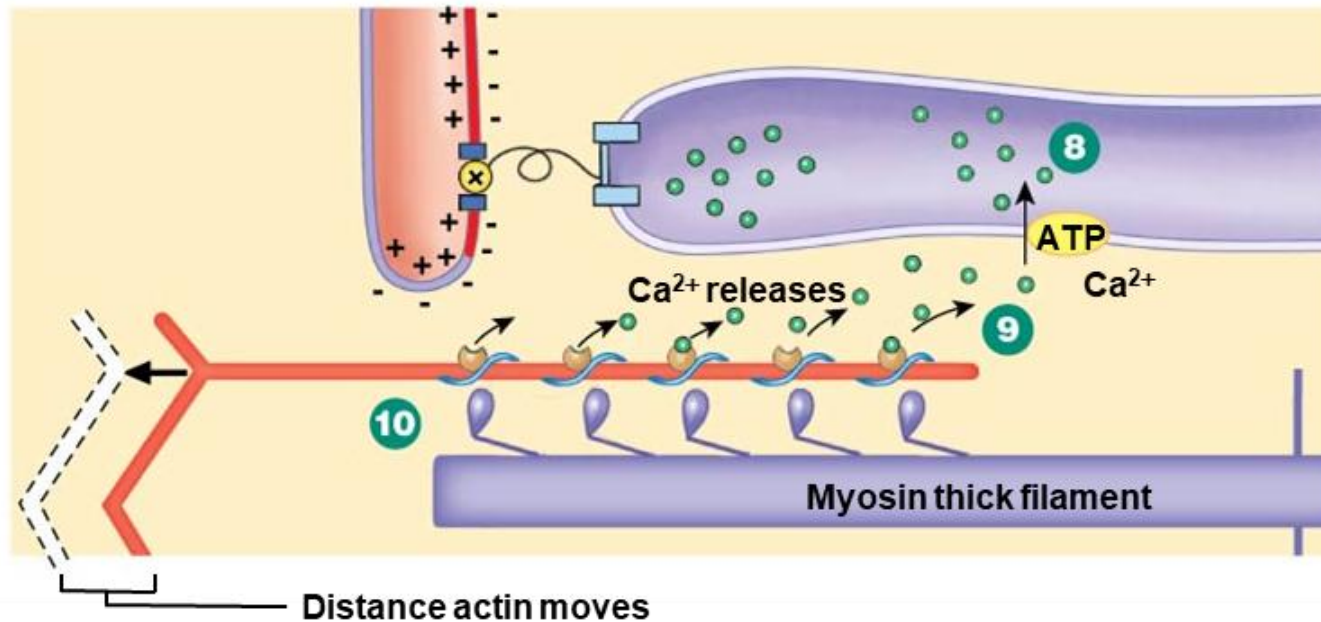
6 Myosin heads execute power stroke.

7 Actin filament slides toward center of sarcomere.

DHP = dihydropyridine L-type calcium channel
RyR = ryanodine receptor-channel

Excitation-Contraction Coupling and Relaxation

(c) Relaxation Phase



8 Sarcoplasmic Ca^{2+} -ATPase pumps Ca^{2+} back into SR.

9 Decrease in free cytosolic $[\text{Ca}^{2+}]$ causes Ca^{2+} to unbind from troponin.

10 Tropomyosin re-covers binding site. When myosin heads release, elastic elements pull filaments back to their relaxed position.

Excitation-Contraction Coupling Summary

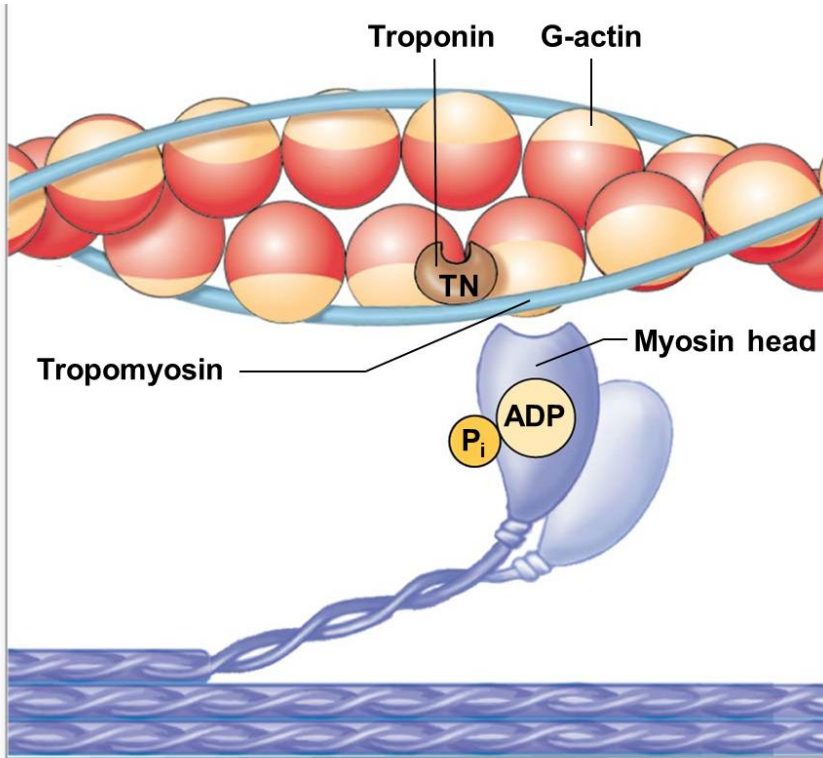
- **Depolarization** of the muscle fiber membrane creates an end-plate potential (**EPP**)
- A muscle **twitch** is a single **contraction-relaxation** cycle
- A **latent period** is the short **delay between the muscle action potential and beginning of muscle tension** development (discussed in MSK Physiology, lec 2)
 - Time required for **calcium release and binding to tropoin**

Clinical Correlation Excitation Contraction Coupling

Malignant hyperthermia

- Mutations or dysregulation of Ca^{2+} release channels in muscle (ryanodine receptors).
- Patients with this condition display normal muscle function under normal conditions.
- However, certain **anesthetic agents** or in rare cases exposure to high environmental heat or strenuous exercise can trigger an abnormal release of Ca^{2+} from the sarcoplasmic reticulum in the muscle cell, resulting in **sustained muscle contraction** and **heat production**.
- Treatment: Dantrolene directly interferes with muscle contraction by binding RyR receptors.

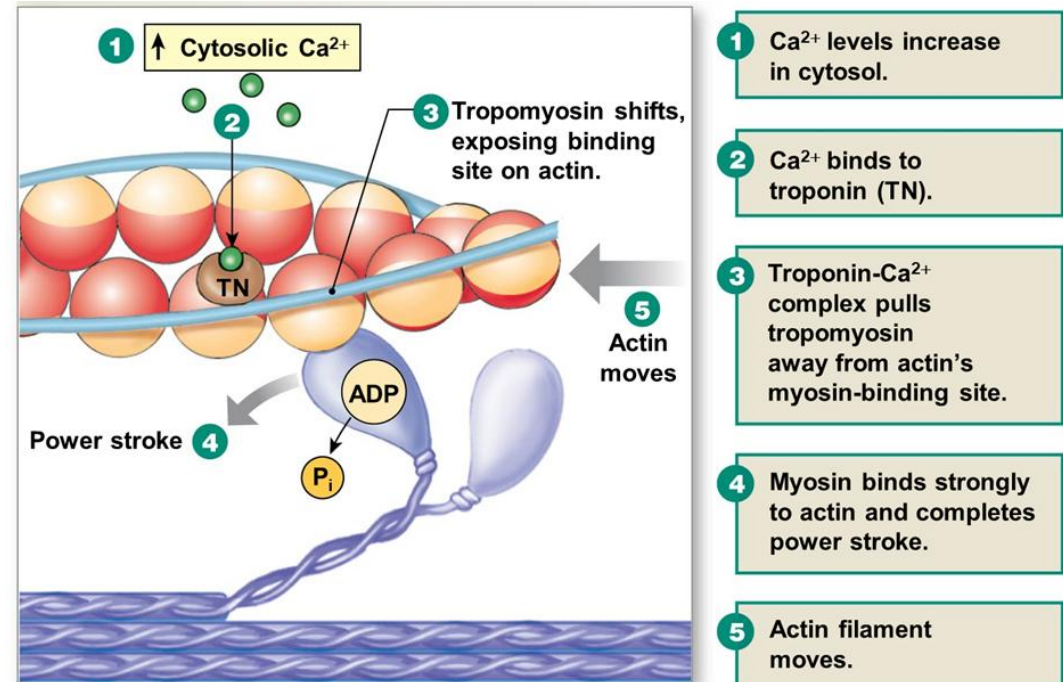
(a) Relaxed state



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- Myosin head cocked
- Tropomyosin partially blocks binding site on actin
- Myosin is weakly bound to actin

(b) Initiation of contraction. A calcium signal initiates contraction



1 Ca^{2+} levels increase in cytosol.

2 Ca^{2+} binds to troponin (TN).

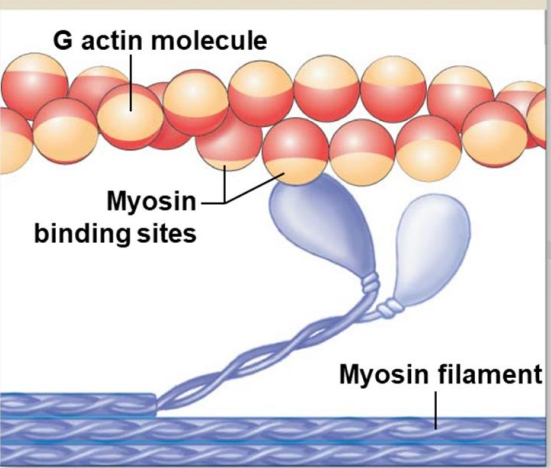
3 Troponin- Ca^{2+} complex pulls tropomyosin away from actin's myosin-binding site.

4 Myosin binds strongly to actin and completes power stroke.

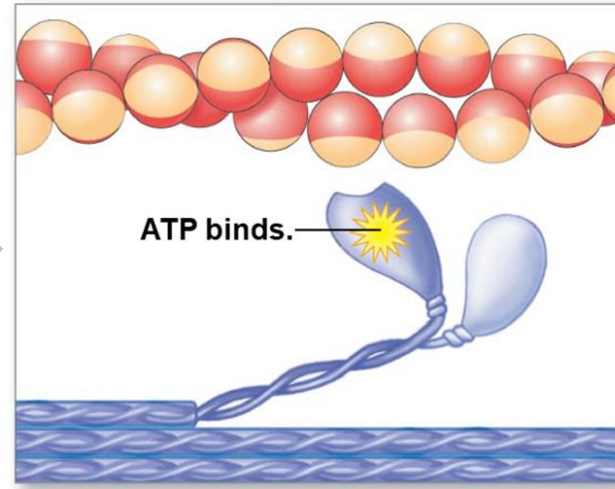
5 Actin filament moves.

The Contraction / Crossbridge Cycle

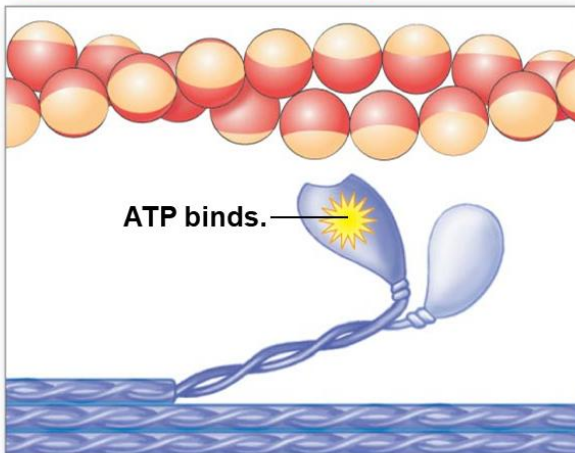
Tight Binding in the Rigor State



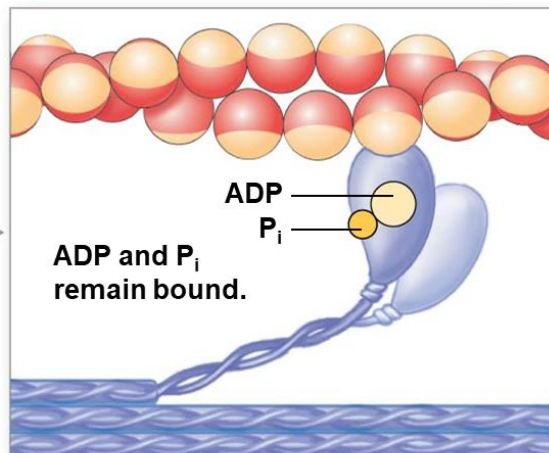
1 ATP binds to myosin.
Myosin releases actin



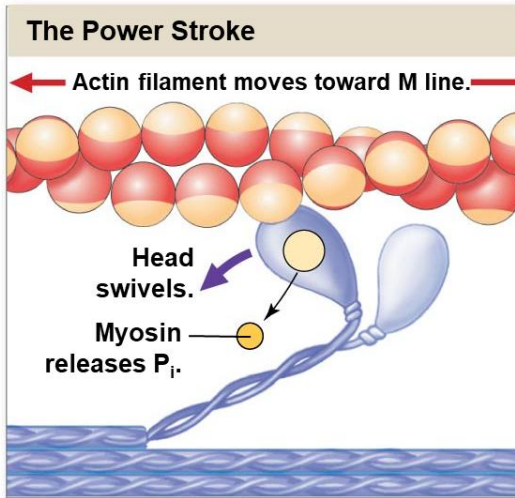
1 ATP binds to myosin.
Myosin releases actin.



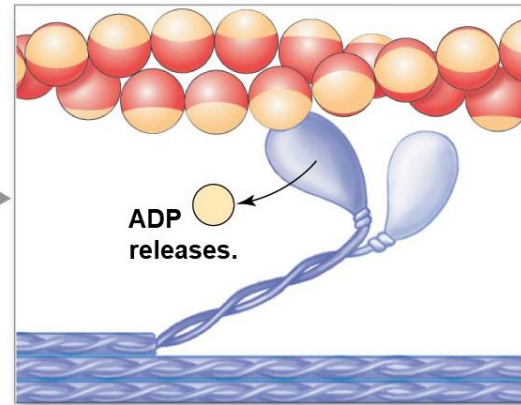
2 Myosin hydrolyzes ATP. Energy from ATP rotates the myosin head to the cocked position. Myosin binds weakly to actin.



- 3 Power stroke begins when tropomyosin moves off the binding site.



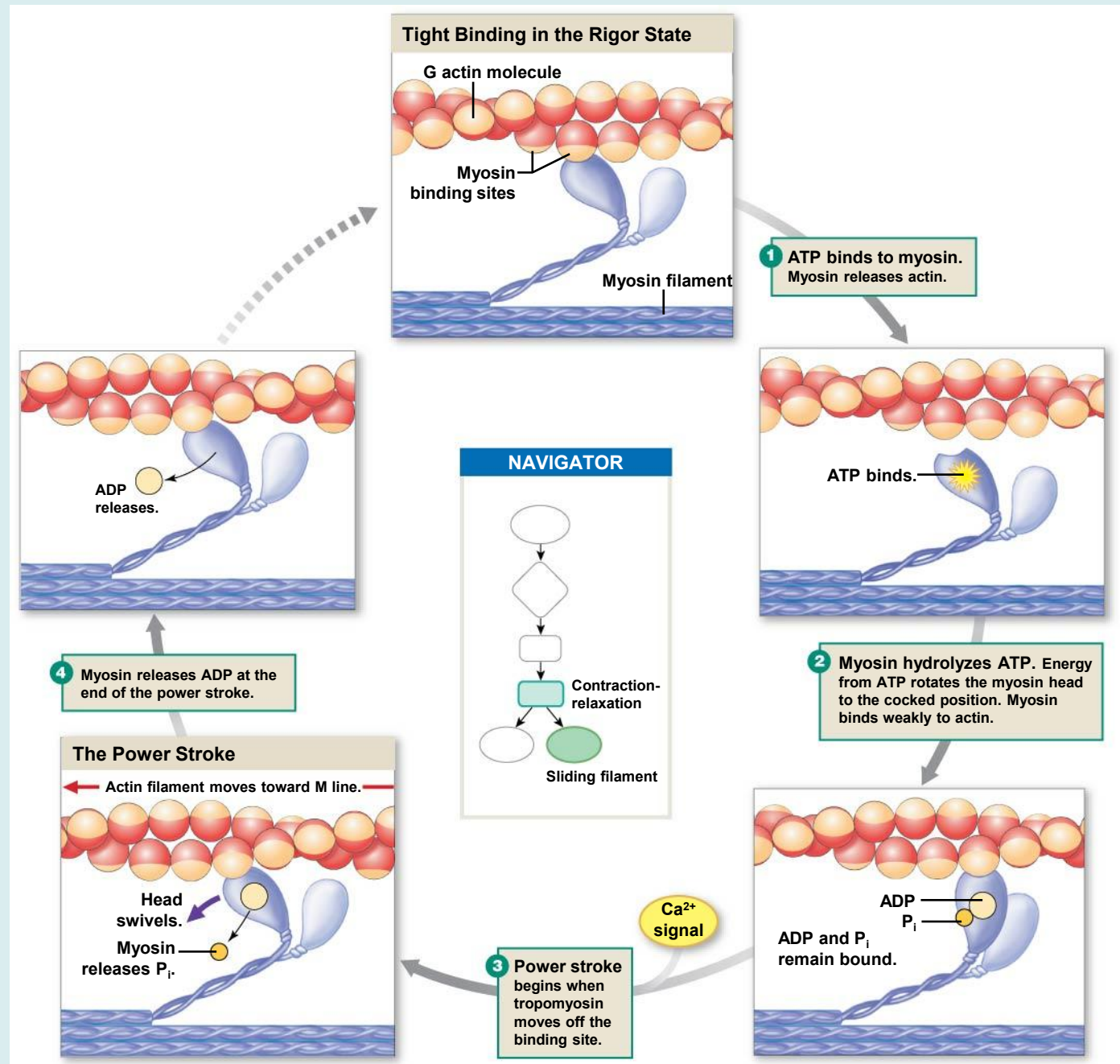
- 4 Myosin releases ADP at the end of the power stroke.



Rigor Mortis

- This cycle is dependent on ATP supply and activation via Ca^{2+} .
- After death cellular energy stores are depleted, detachment can't occur because of the lack of ATP, and the cycle stops in an attached state.
- This produces an overall stiffness of the muscle, which is observed as the Rigor Mortis that sets in shortly after death.

Summary Contraction / Crossbridge Cycle



SOM.1a.BPM1.2.MSK.2.PHYS.0054

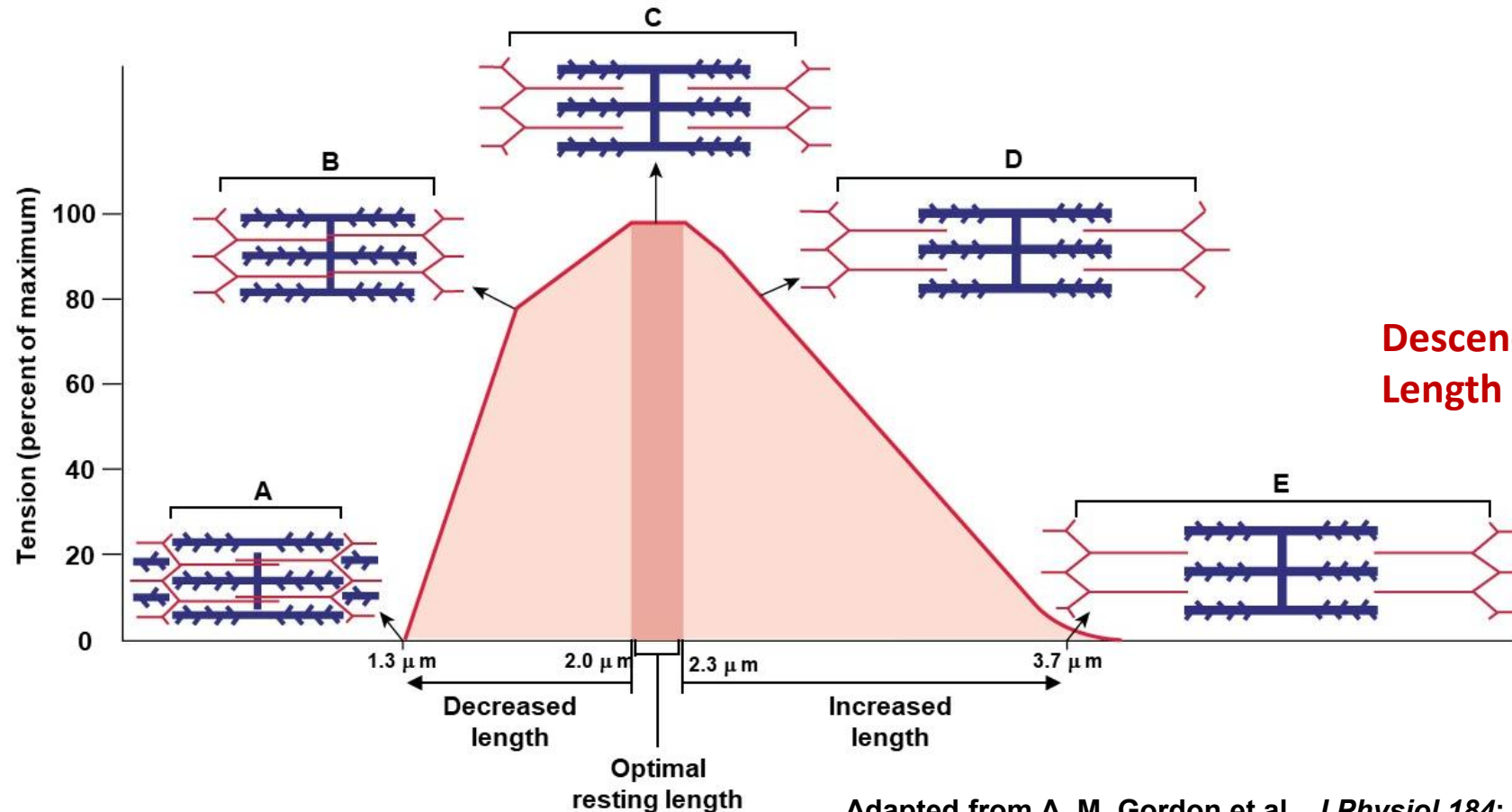
SOM.1a.BPM1.2.MSK.2.PHYS.0057

Length-Tension Relationship

Sarcomere length and the amount of myosin-actin filament overlap leads to **active tension** development by a **contracting muscle fiber**. Too much or too little overlap of thick and thin filaments in resting muscle = **Decreased tension**

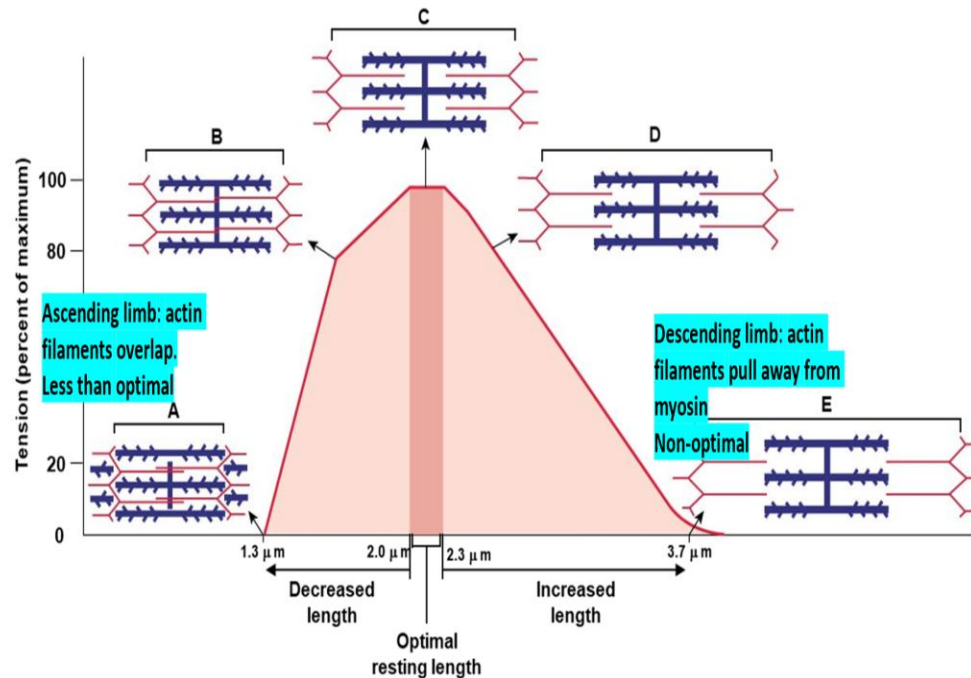
Ascending curve:
 $\text{Length} \propto \text{Force}$

Descending curve:
 $\text{Length} \propto 1/\text{Force}$



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Adapted from A. M. Gordon et al., *J Physiol* 184: 170–192, 1966.



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SOM.1a.BPM1.2.MSK.2.PHYS.0063

Which point of the graph indicates the muscle is overstretched?

- Point A, actin filaments are overlapping each other more than they are overlapping myosin, thus less crossbridge formation (power stroke). The amount of active tension developed in the muscle is far less than maximum.
- Point B actin filaments are overlapping each other less, thus they are overlapping myosin filaments more which leads to more crossbridge formation (power stroke) and more active tension being developed.
- Point C actin filaments are completely overlapping the myosin filaments allowing for maximum crossbridge formation which leads to maximum tension development.
- Point D actin filaments are pulling away from myosin filaments, thus less crossbridge formation, thus active tension decreases.
- Point E actin filaments have nearly completely pulled away from myosin filaments, thus far less crossbridge formation, thus active tension is decreased further.
- Increase actin-myosin overlap → more crossbridge formation → more active tension development.
- In vivo muscles are at point C

References

- Ganong's Medical Physiology Examination & Board Review
- Kaplan USMLE STEP 1 Lecture Notes 2016
- Silverthorn human physiology, 6th edition
- Principles of Human Physiology by Stanfield and Germann, 4th edition